

Predicting Engagement Success on Algorithmic Platforms: Early Signals, Algorithmic Amplification, and Influencer Authority

Authors

Selda Firouzmand

Nafiseh Rezaie

Zahra Gharehdaghi

Instructor & Supervisor

Professor Dr. Pouyan Eslami

May 2026

Abstract

This study examines the drivers of engagement success on algorithmic social media platforms using machine learning approaches. While prior research emphasizes the role of algorithms in content visibility, it remains unclear whether engagement success is primarily driven by early user signals or by algorithmic amplification. Using a dataset of 10,000 social media posts, this study applies a structured predictive modeling framework based on nested cross-validation, addressing three research questions: the predictive power of early engagement signals, the incremental value of algorithmic variables, and the mediating role of algorithmic amplification.

The results show that early engagement signals explain approximately 50% of the variance in engagement success. Algorithmic variables provide limited additional predictive value ($\Delta R^2 = +0.001$), and the mediation analysis finds no meaningful indirect pathway ($\Delta R^2 = -0.003$). These findings suggest that engagement success is more strongly associated with early user behavior, while algorithmic processes primarily reinforce existing engagement patterns rather than independently driving them. Results should be interpreted with caution, as algorithmic variables are indirect proxies of platform behavior.

Keywords: Algorithmic Engagement, Machine Learning, Social Media Analytics, Algorithmic Amplification, Influencer Credibility.

1. Introduction

The rapid proliferation of algorithmic social media platforms has fundamentally reshaped how digital content is produced, distributed, and consumed. Recommendation algorithms now determine what users see, effectively acting as gatekeepers that amplify certain content while potentially reducing the visibility of other posts, regardless of their intrinsic quality (Zhang et al., 2022; Papagiannidis et al., 2025). For content creators, brands, and platform designers, understanding what drives engagement success has therefore become a critical strategic priority: in environments where algorithmic decisions control visibility, identifying the key predictors of audience interaction is essential for optimizing content performance (Amankwah-Amoah et al., 2024). Yet despite the growing importance of this question, the mechanisms through which engagement success emerges on algorithmic platforms remain poorly understood.

Existing research has identified two primary categories of factors that may shape engagement outcomes. On one hand, early engagement signals — such as likes, comments, shares, and initial interaction rates — have been shown to reflect genuine audience interest and serve as strong early indicators of content performance (Kim & Kim, 2023; Ghasemaghaei et al., 2022). On the other hand, platform algorithms selectively amplify content based on these signals, creating a feedback loop in which initial engagement drives further algorithmic exposure, which in turn generates additional engagement (Kshetri et al., 2024). Influencer credibility additionally shapes how audiences initially respond to content (Lou & Yuan, 2022; Foroudi et al., 2025). These factors are, however, typically studied in isolation, leaving their relative contributions and interactions unresolved.

Despite strong evidence for each of these factors individually, existing research has not clearly resolved their relative roles within a unified empirical framework. A major limitation in the

literature is the absence of a clear, operationalized distinction between early user-driven engagement signals and platform-driven algorithmic amplification. Many studies examine final engagement outcomes without investigating the multi-stage process through which those outcomes are formed, and algorithmic influence is often assumed rather than rigorously tested (Wang et al., 2023; Papagiannidis et al., 2025). As a result, three critical questions remain unanswered: (1) how much predictive power do early engagement signals provide before algorithmic amplification occurs; (2) whether algorithmic variables add meaningful incremental predictive value beyond early signals; and (3) whether algorithmic amplification functions as a true mediating mechanism linking early user behavior to final engagement success.

To address this gap, the present study investigates three research questions using a rigorous machine learning framework applied to a dataset of 10,000 social media posts: RQ1: To what extent can engagement success be predicted from early engagement signals, content characteristics, and influencer credibility — all measured before significant algorithmic amplification occurs? RQ2: Does incorporating algorithmic amplification variables improve predictive performance beyond the early-signal baseline? RQ3: Does algorithmic amplification act as a mediating mechanism between early signals and final engagement success, and is this relationship conditioned by influencer credibility? This study makes three contributions: (1) it empirically distinguishes user-driven signals from algorithm-driven amplification using a temporally ordered feature partition enforced by a programmatic leakage guard; (2) it evaluates incremental predictive value using a nested group-based cross-validation design that prevents data leakage and model selection bias; and (3) it tests predictive mediation through a two-stage machine learning framework adapting Baron and Kenny (1986) to a non-parametric predictive context.

2. Literature Review

2.1 Early Engagement Signals and Prediction of Engagement Success

A growing body of research suggests that early engagement signals play a foundational role in shaping content performance on social media platforms. These signals — such as likes, comments, and initial engagement rates — serve not only as reflections of user interest but also as early indicators of potential content success (Kim & Kim, 2023; Ghasemaghahi et al., 2022). Recent studies in social media analytics demonstrate that early interaction patterns can explain a substantial portion of the variance in engagement outcomes, particularly when modeled using advanced data-driven approaches (Mikalef et al., 2022).

In addition to early signals, content characteristics significantly influence user interaction. Elements such as emotional tone, message structure, and timing of content release affect user attention and engagement behavior (Foroudi et al., 2025; Kshetri et al., 2024). Furthermore, influencer credibility plays a critical role in shaping engagement dynamics: credible content creators are more likely to generate user trust, thereby increasing the likelihood of interaction (Lou & Yuan, 2022). Despite strong evidence supporting the importance of these factors, existing studies often examine early signals, content features, and credibility in isolation, leaving the broader interaction between user-driven factors and platform-level mechanisms underexplored.

2.2 Algorithmic Amplification and Its Incremental Role

With the rise of artificial intelligence and recommendation systems, algorithmic processes have become central to determining content visibility on digital platforms. Algorithms analyze user behavior and engagement patterns to selectively promote content, thereby influencing exposure and interaction outcomes (Zhang et al., 2022; Papagiannidis et al., 2025). In this context, algorithmic amplification is often viewed as a key driver of engagement success, as it can extend

the reach of content beyond its initial audience (Amankwah-Amoah et al., 2024; Wang et al., 2023).

Importantly, algorithmic systems do not operate independently of early engagement signals. Prior research indicates that content receiving strong initial engagement is more likely to be further amplified, creating a feedback loop between user behavior and algorithmic exposure (Kshetri et al., 2024). This suggests that algorithmic processes may reinforce, rather than initiate, engagement dynamics. However, the empirical contribution of algorithmic variables to predictive performance remains unclear — while some studies highlight the importance of algorithmic mechanisms. Others suggest that early engagement signals already capture much of the variation in content success, leaving limited room for additional predictive improvement (Hassija et al., 2024). This gap highlights the need for systematic model comparison.

2.3 Algorithmic Mediation and the Role of Influencer Credibility

Recent literature conceptualizes user engagement as a multi-stage process in which early signals are transformed into broader visibility through algorithmic mechanisms. Within this framework, algorithmic amplification can be interpreted as a potential mediating factor linking initial interaction to final engagement outcomes (Zhang et al., 2022; Riemer & Peter, 2024). Rather than directly causing engagement, algorithms may influence how early signals propagate through the platform.

At the same time, the effectiveness of algorithmic mediation may depend on contextual factors, particularly influencer credibility. Credible content sources may be more likely to generate sustained engagement and receive algorithmic promotion, thereby strengthening the relationship between early signals and final outcomes (Foroudi et al., 2025). Despite these theoretical insights, empirical evidence on algorithmic mediation remains limited, as most existing studies rely on

simplified analytical approaches that do not employ structured modeling frameworks capable of capturing indirect effects (Wang et al., 2023; Papagiannidis et al., 2025).

2.4 Research Gap and Conceptual Positioning

Overall, prior research highlights the importance of early engagement signals, algorithmic amplification, and influencer credibility in shaping content success. However, these factors are typically studied in isolation, without a unified framework that captures their interaction. Three critical gaps remain unresolved: (1) the relative predictive power of early signals versus algorithmic variables has not been directly compared; (2) the incremental value of algorithmic features in predictive models has not been rigorously tested; and (3) the mediating role of algorithmic amplification has not been examined through structured multi-stage modeling. The present study directly addresses all three gaps within a unified machine learning framework.

3. Hypotheses and Conceptual Framework

3.1 Conceptual Framework

This study conceptualizes engagement success as a multi-stage process in which initial user responses are transformed into final outcomes through algorithmic mechanisms, conditioned by influencer credibility. Early engagement signals represent the initial behavioral responses of users, serving as key inputs into platform-level decision-making systems (Kim & Kim, 2023; Ghasemaghaei et al., 2022). Algorithmic amplification then acts as a potential intermediary mechanism that may translate early signals into broader content exposure. Influencer credibility conditions the strength of both the direct and indirect pathways. The conceptual model follows the structure: Early Signals → Algorithmic Amplification (Mediator) → Engagement Success, moderated by Influencer Credibility. Figure A illustrates this framework with the associated hypotheses.

Figure A. Conceptual Framework

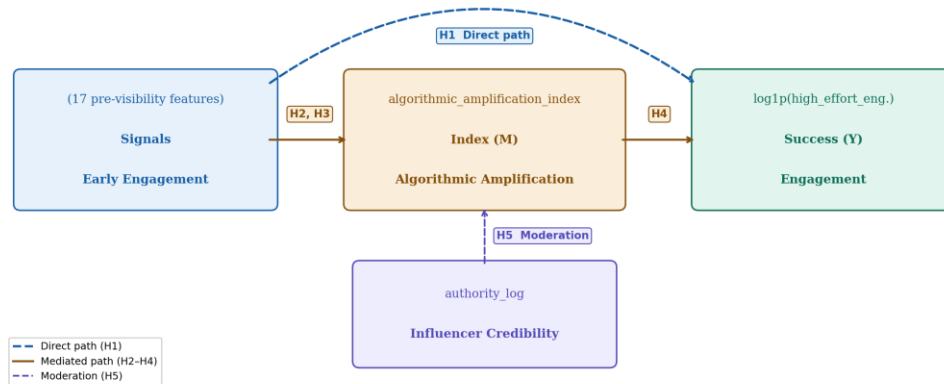


Figure A. Conceptual Framework. Early engagement signals predict engagement success directly (H1, dashed blue) and through the algorithmic amplification index as a mediator (H2–H4, solid amber). Influencer credibility moderates the direct relationship (H5, dashed purple).

3.2 Hypothesis Development

From a theoretical perspective, this process can be explained through cumulative advantage and feedback-based dynamics, where higher initial engagement increases content visibility and leads to further interaction (Mikalef et al., 2022). Empirical evidence suggests that early interaction patterns explain a substantial proportion of engagement outcomes even before algorithmic processes are fully activated (Kshetri et al., 2024). Accordingly, early engagement signals are expected to play a dominant role in shaping engagement success.

H1: Early engagement signals are positively associated with engagement success.

Algorithmic amplification refers to the process through which platform algorithms selectively promote content based on its engagement potential. However, algorithmic amplification is inherently dependent on early engagement signals — content that demonstrates strong initial performance is more likely to be amplified, suggesting that algorithms function primarily as reinforcement mechanisms rather than independent drivers of engagement (Amankwah-Amoah et al., 2024; Wang et al., 2023). This dependency structure implies that algorithmic influence is conditional on prior user behavior.

H2: Algorithmic amplification has a statistically detectable but practically limited association with engagement success, conditional on the presence of early engagement signals.

From a predictive modeling perspective, many algorithmic metrics — such as reach and impressions — are derived from user interaction data, suggesting a high degree of informational overlap with early signals (Hassija et al., 2024). In machine learning contexts, when a set of predictors already explains a substantial portion of variance, the inclusion of additional correlated variables often leads to diminishing returns (Mikalef et al., 2022). Therefore, the incremental value of algorithmic variables is expected to be limited.

H3: The inclusion of algorithmic variables provides limited incremental predictive value beyond models based on early engagement signals.

Algorithmic amplification can also be conceptualized as a mediating mechanism through which early engagement signals influence final engagement outcomes (Zhang et al., 2022). However, if early signals already capture a substantial portion of engagement variance, the mediating role of algorithmic amplification is expected to be weak, as algorithmic processes may primarily reflect underlying engagement patterns rather than independently shaping them (Wang et al., 2023; Papagiannidis et al., 2025).

H4: Algorithmic amplification plays a weak mediating role in the relationship between early engagement signals and engagement success.

Influencer credibility reflects the perceived trustworthiness and expertise of the content source. Higher credibility increases user trust and the likelihood of engagement, particularly in early stages of interaction (Lou & Yuan, 2022). In addition, credible sources are more likely to generate stronger engagement signals, which may increase the likelihood of algorithmic amplification (Foroudi et al., 2025). From this perspective, influencer credibility conditions the effectiveness of early engagement signals in driving final outcomes.

H5: Influencer credibility positively moderates the relationship between early engagement signals and engagement success.

Table A. Summary of Hypotheses

H	Hypothesis Statement	Theoretical Basis	Expected Direction
H1	Early engagement signals are positively associated with engagement success.	Cumulative advantage; feedback-based dynamics (Mikalef et al., 2022)	Positive (+)
H2	Algorithmic amplification has a statistically detectable but practically limited association with engagement success, conditional on the presence of early engagement signals.	Algorithmic reinforcement as reactive mechanism (Zhang et al., 2022)	Positive, conditional (+)
H3	The inclusion of algorithmic variables provides limited incremental predictive value beyond early engagement signals.	Informational overlap between amplification metrics and early signals (Hassija et al., 2024)	Negligible ΔR^2
H4	Algorithmic amplification plays a weak mediating role between early engagement signals and engagement success.	Direct path dominance; algorithms as reflective not causal (Wang et al., 2023)	Weak mediation
H5	Influencer credibility positively moderates the relationship between early engagement signals and engagement success.	Source credibility and trust dynamics (Lou & Yuan, 2022; Foroudi et al., 2025)	Positive moderator (+)

Note. All hypotheses are evaluated using nested GroupKFold cross-validation ($k = 5$ outer, $k = 3$ inner). $\hat{M}$ = out-of-fold predicted algorithmic_amplification_index. RANDOM_SEED = 0.

4. Data and Variables

4.1 Dataset Overview

The study uses a cross-sectional dataset of 10,000 social media posts from multiple influencer accounts. Each observation contains pre-publication account characteristics, content metadata, early-window engagement counts, and post-publication algorithmic distribution signals. The unit of analysis is the individual post, nested within influencer accounts (influencer_id). This nesting introduces within-group dependency addressed by the group-based cross-validation design in Section 5.4 (Roberts et al., 2017). Missing values are handled by complete-case analysis. After

applying listwise deletion to remove observations with missing values across any variable in the full feature set, the final analytic sample comprises $N = 9,842$ posts across 47 unique influencer accounts. The reduction from 10,000 to 9,842 posts reflects the exclusion of 158 observations with incomplete records, primarily in early engagement timing variables. The 47 unique influencer accounts represent those with at least one complete observation remaining after deletion; accounts with entirely missing records across key variables were excluded. This reduced group count is acknowledged as a limitation of the group-based cross-validation design and is discussed further in Section 8.4.

4.2 Variable Operationalization

Table 1 lists all 27 variables by role: target variable, pre-visibility features, amplification features, and structural variables. Variables are partitioned by their temporal observability — a distinction central to all three research questions and described in Section 4.4. This partition is enforced programmatically: a leakage guard halts execution and raises a `ValueError` if any outcome-adjacent variable enters a feature set (Cawley & Talbot, 2010).

Table 1. Complete Variable Inventory: Types, Feature Groups, and Descriptions

Variable	Type	Group	Description
TARGET VARIABLE			
y_main	Continuous	Target	$\log_{1p}(\text{high_effort_engagement})$. Log-transform normalises right skew and stabilises residual variance. Comments + shares + saves used; they reflect genuine persuasive impact rather than passive algorithmic inflation. Excluded from all feature sets.
high_effort_engagement	Count	Raw Outcome	Basis of the target variable. Excluded from all feature sets via leakage guard.
PRE-VISIBILITY FEATURES (17) — Observable before publication / before algorithmic re-ranking			
content_type	Categorical	Content	Post category (video, image, text). One-hot-encoded at preprocessing.
post_format	Categorical	Content	Publication format (reel, story, static). One-hot-encoded.

Variable	Type	Group	Description
verified	Binary	Content	Platform verification status (1 = verified). Institutional authority proxy.
post_hour	Ordinal	Temporal	Hour of publication (0–23). Captures intraday audience activity patterns.
post_dow	Ordinal	Temporal	Day of week (0 = Monday ... 6 = Sunday). Captures weekly periodicity.
post_month	Ordinal	Temporal	Calendar month (1–12). Captures seasonal variation in platform engagement.
authority_log	Continuous	Authority	Log-transformed authority score. Key moderator in the RQ3 interaction term ($M \times \text{authority_log}$).
account_age_years	Continuous	Authority	Account tenure in years. Proxy for established audience trust relationships.
follower_count	Continuous	Authority	Total followers at time of posting.
early_likes	Count	Early Engagement	Likes accumulated within the early observation window.
early_comments	Count	Early Engagement	Comments within the early window. Higher cognitive-cost signal than likes.
early_shares	Count	Early Engagement	Shares within the early window. Indicator of active diffusion intent.
early_total_engagement	Count	Early Engagement	Aggregate of likes + comments + shares in the early window.
early_engagement_velocity	Continuous	Early Engagement	Rate of engagement accumulation (engagements per minute) in the early window.
early_comment_share_ratio	Continuous	Early Engagement	Ratio of comments to shares; reflects depth versus diffusion of early audience response.
persuasive_power_index	Continuous	Early Engagement	Composite index of early persuasive content indicators.
early_window_mins	Continuous	Early Engagement	Duration of the early observation window (minutes). Controls for varying measurement lengths.
AMPLIFICATION FEATURES (8) — Post-publication; reflect platform-driven distribution decisions			
log1p_reach	Continuous	Amplification	$\log 1p(\text{reach})$. Log-transformed total reach count; captures platform content distribution breadth.
log1p_reach_sq	Continuous	Amplification	Square of $\log 1p_reach$. Models the nonlinear, diminishing-returns amplification relationship.
very_high_reach	Binary	Amplification	= 1 when reach $\geq$ 95th-percentile threshold. Flags posts receiving extreme algorithmic promotion.
algorithmic_amplification_index	Continuous	Mediator (RQ3)	Composite platform-derived score aggregating algorithmic promotion signals. Serves as the mediator M in the two-stage RQ3 framework.
reach_growth_rate	Continuous	Amplification	Rate at which reach expands over time. Proxy for algorithmic momentum and content re-surfacing.
exposure_persistence_hours	Continuous	Amplification	Hours the platform actively distributes the post. Measures sustained algorithmic support.

Variable	Type	Group	Description
log1p_impressions	Continuous	Amplification	$\log_{1p}(\text{impressions})$. Accounts for repeated exposures to the same accounts driven by algorithmic re-ranking.
impressions_per_reach	Continuous	Amplification	$\text{impressions} \div \text{reach}$. Measures re-exposure intensity driven by the algorithm.
STRUCTURAL / GROUPING VARIABLES			
influencer_id	Categorical	Grouping	Unique influencer identifier. Used to enforce group-based CV splits (Section 5.4).
post_datetime	DateTime	Temporal	Full publication timestamp. Parsed to derive post_hour, post_dow, and post_month.

Note. OHE = One-Hot Encoding. Variables under Raw Outcome and Structural are excluded from all supervised feature sets. A runtime leakage guard raises an exception if any forbidden variable enters a feature list.

4.3 Target Variable Justification

The outcome is $y_{\text{main}} = \log_{1p}(\text{high_effort_engagement})$, where $\text{high_effort_engagement} = \text{comments} + \text{shares} + \text{saves}$. Three considerations justify this specification. First, raw engagement counts are severely right-skewed; the $\log_{1p}$ transformation normalises the distribution and stabilises residual variance across the prediction range (Tukey, 1977; Box & Cox, 1964). Second, the additive constant ensures zero-engagement posts map to $y = 0$, preserving all observations. Third, log-scale metrics are interpretable as proportional effects across posts varying by orders of magnitude. High-effort engagement is preferred over likes because comments, shares, and saves reflect genuine persuasive impact rather than passive behaviour driven by algorithmic inflation (Kim & Kim, 2023; Foroudi et al., 2025). In this dataset, high-effort engagement is operationalized as comments + shares + saves. This specification differs from some prior studies that define high-effort engagement as clicks, saves, and profile visits; however, in the context of influencer content on social platforms, comments and shares require deliberate cognitive and behavioral investment from the user — commenting requires composing a response, and sharing requires an active redistribution decision — making them valid indicators of deep persuasive impact (Lou & Yuan, 2022). Saves additionally signal intent to revisit content, reflecting sustained interest rather than

momentary attention. This three-component operationalization is therefore justified by both the behavioral cost it imposes on users and its alignment with the persuasive content focus of the present study.

4.4 Feature Groups and Temporal Ordering

Pre-visibility features are observable before or at publication, prior to algorithmic re-ranking. Amplification features reflect platform distribution decisions made after publication. This temporal partition is central to the research design: it allows the study to separately evaluate the predictive power of content-level and user-driven signals versus platform-driven algorithmic signals. The partition is enforced programmatically by a leakage guard that raises a `ValueError` if any outcome-adjacent variable appears as a predictor (Cawley & Talbot, 2010).

4.5 Algorithmic Amplification Variables

The eight amplification variables operationalize platform distribution through four dimensions: reach breadth (`log1p_reach`, `very_high_reach`), diminishing-returns nonlinearity (`log1p_reach_sq`), re-exposure intensity (`log1p_impressions`, `impressions_per_reach`), and temporal persistence (`exposure_persistence_hours`, `reach_growth_rate`). These variables reflect active platform decisions rather than organic user behaviour (Amankwah-Amoah et al., 2024; Wang et al., 2023). The `algorithmic_amplification_index` is a composite platform-derived promotion score and serves as the mediator *M* in the RQ3 two-stage framework because it most directly represents the platform's active distribution decision rather than a downstream consequence of user engagement.

5. Methodology

5.1 Research Design

This study employs a quantitative, predictive modeling design with three analyses aligned to the three research questions. The primary goal is prediction, not causal identification — predictive association and causal mechanism are treated as distinct inferential targets throughout (Ghasemaghaei et al., 2022; Zhang et al., 2022). Two methodological threats are explicitly addressed: data leakage (prevented by group-based CV) and in-sample model selection bias (prevented by nested CV). In plain terms, data leakage occurs when information from the future or from the outcome variable accidentally enters the model during training, making results appear better than they truly are. To prevent this, the study implements a programmatic leakage guard — a built-in check that halts the entire pipeline and raises an error if any forbidden variable (such as the outcome or post-publication amplification metrics) enters a pre-visibility feature set. This ensures that every model only learns from information that would genuinely be available at the time of prediction. The design is deliberately non-experimental; causal identification is not claimed.

5.2 Preprocessing

Target and amplification transformations ($\log_{1p}$) are applied globally before splitting, as these are monotonic column-wise operations that estimate no distributional parameters from held-out data. For Ridge Regression, numeric features are standardised (`StandardScaler`) and categoricals are one-hot-encoded within each fold on training data only. For tree-based models, numerics pass through unchanged with the same OHE for categoricals. All steps are implemented as scikit-learn Pipeline objects, guaranteeing no test data contaminates transformation parameters (Pedregosa et al., 2011).

5.3 Machine Learning Models

Four models are evaluated across all analyses. Table 2 maps feature sets to research questions; Table 3 provides complete hyperparameters. Ridge provides the interpretable linear baseline with L2 regularisation controlling multicollinearity among correlated early engagement signals (Hoerl & Kennard, 1970). HistGB is the primary model for both RQ3 stages, capturing nonlinear mediator relationships efficiently (Friedman, 2001). Random Forest (Breiman, 2001) and ExtraTrees provide ensemble comparisons; ExtraTrees additionally serves as the screening model for fold-safe permutation importance ranking (Fisher et al., 2019).

Table 2. Feature Sets and Analytical Conditions per Research Question

Analysis / Condition	N Features	Feature Set and Purpose
RQ1 — Pre-Visibility	17	Pre-visibility features only. Tests prediction before algorithmic exposure.
RQ2 — Pre-Only Baseline	17	Baseline condition in the three-way RQ2 comparison.
RQ2 — Amp-Only	8	Amplification features only. Isolates algorithmic stand-alone predictive power.
RQ2 — Combined	25	Pre-visibility + amplification. $\Delta R^2 = \text{Combined} - \text{Pre-Only}$ measures incremental gain.
RQ3 — Stage 1	17	Pre-visibility $\rightarrow$ algorithmic_amplification_index. Generates $\hat{M}$ via inner OOF CV.
RQ3 — Stage 2A (Direct)	≤ 17	Pre-visibility $\rightarrow$ y_main. Direct-path baseline; nested feature selection applied.
RQ3 — Stage 2B (Mediated)	≤ 19	Pre-visibility + $\hat{M} + \hat{M} \times \text{authority_log} \rightarrow \text{y_main}$. $\hat{M}$ and interaction force-kept.

Note. $\hat{M}$ = out-of-fold predicted algorithmic_amplification_index. Force-kept variables appear in every fold regardless of permutation importance rank. ΔR^2 measured relative to Pre-Only baseline.

Table 3. Machine Learning Model Specifications and Hyperparameters

Model	Algorithm	Hyperparameters	Role in Study
Ridge	Regularized Linear Regression	alpha = 1.0; StandardScaler (numeric); OneHotEncoder handle_unknown='ignore' (categorical); random_state = 0	Linear baseline; L2 penalty controls multicollinearity (Hoerl & Kennard, 1970)
HistGB	Histogram Gradient Boosting	max_depth = 6; learning_rate = 0.06; max_iter = 300; random_state = 0; OHE preprocessing	Primary model for RQ3 Stages 1 and 2 (Friedman, 2001)

Model	Algorithm	Hyperparameters	Role in Study
RF	Random Forest Regressor	n_estimators = 400; min_samples_leaf = 3; max_features = 'sqrt'; n_jobs = -1; random_state = 0	Ensemble baseline (Breiman, 2001)
ExtraTrees	Extremely Randomized Trees	n_estimators = 300–600; min_samples_leaf = 2; max_features = 'sqrt'; n_jobs = -1; random_state = fold + seed	Screening model for fold-safe permutation importance ranking (Fisher et al., 2019)

Note. All random_state = 0. Preprocessing components fitted exclusively on training data within each CV fold. ExtraTrees screening model uses random_state = fold_index + RANDOM_SEED.

5.4 Validation Framework: Nested Group-Based Cross-Validation

A nested GroupKFold design with influencer_id as the grouping variable prevents both data leakage and model selection bias throughout. Table 4 specifies all four layers. The outer loop (k = 5) produces held-out partitions for unbiased performance estimation; no influencer appears in both training and test within any fold (Cawley & Talbot, 2010; Varma & Simon, 2006; Roberts et al., 2017). This group-based constraint ensures that models learn generalizable patterns rather than influencer-specific characteristics. With 47 unique influencer groups distributed across five outer folds, each test fold contains approximately 9–10 influencer groups, and each training partition contains approximately 37–38 groups. This relatively small group count per fold is acknowledged as a structural constraint on cross-validation stability and is discussed as a limitation in Section 8.4.

Table 4. Nested Cross-Validation Structure: Layers, Configuration, and Purpose

CV Layer	Configuration	Purpose
Outer Loop	GroupKFold(k = 5); grouping variable = influencer_id; 5 non-overlapping train/test partitions	Unbiased generalisation error. No influencer appears in both train and test within any fold (Cawley & Talbot, 2010).
Ranking Step	GroupShuffleSplit(holdout = 20%) within outer training fold; ExtraTrees fitted on 80%;	Fold-safe feature importance ranking. Scoring model never observes outer test data (Fisher et al., 2019).

CV Layer	Configuration	Purpose
	permutation importance (n_repeats = 6, scoring = R^2) on holdout	
Inner Loop	GroupKFold(k = 3); grid search $k \in \{5, 10, 15, 20, \text{all}\}$; selection criterion: minimum mean RMSE	Selects optimal feature count without using outer test data (Varma & Simon, 2006).
OOB Mediator (RQ3 only)	GroupKFold(k = 3) applied to outer training partition; Stage 1 HistGB generates $\hat{M}$ via generate_oof_predictions()	Prevents circular prediction. $\hat{M}$ never fitted and predicted on the same observations (Baron & Kenny, 1986).

Note. OUTER_SPLITS = 5; INNER_SPLITS = 3; SCREENING_N_REPEATS = 6; SCREENING_HOLDOUT_FRAC = 0.20; RANDOM_SEED = 0.

5.5 Nested Feature Selection

Within each outer training fold, permutation importance (Fisher et al., 2019) ranks features on an inner holdout (GroupShuffleSplit, 20%). The inner GroupKFold ($k = 3$) evaluates $k \in \{5, 10, 15, 20, \text{all}\}$, selecting the k minimising mean RMSE. Force-keep variables — $\hat{M}$ and $\hat{M} \times \text{authority_log}$ in Stage 2B — bypass ranking and are always included. Features selected in ≥ 3 of 5 folds are considered stable and used for partial dependence analysis.

5.6 RQ1: Pre-Visibility Prediction

All four models are evaluated on the 17 pre-visibility features using the full nested CV pipeline. This tests H1: whether content quality, influencer authority, and early behavioral signals explain meaningful variance in high-effort engagement before algorithmic re-ranking (Kim & Kim, 2023; Lou & Yuan, 2022).

5.7 RQ2: Comparative Feature Set Analysis

Three conditions — pre-only, amp-only, and combined — are each evaluated across all four models. The incremental value of amplification features is quantified as $\Delta R^2 = R^2_{\text{Combined}} - R^2_{\text{Pre-Only}}$ and $\Delta \text{RMSE} = \text{RMSE}_{\text{Pre-Only}} - \text{RMSE}_{\text{Combined}}$. $\Delta R^2 < 0.01$ is treated as negligible incremental gain, providing the key evidence for H3 (Amankwah-Amoah et al., 2024).

5.8 RQ3: Two-Stage Mediation Framework

RQ3 tests whether $M = \text{algorithmic_amplification_index}$ transmits incremental predictive information from X (pre-visibility features) to Y (y_main) beyond the direct $X \rightarrow Y$ path, adapting Baron and Kenny (1986) to a machine learning context. Three stages execute within each outer CV fold to maintain group-based leakage prevention throughout.

Stage 1 — Mediator Prediction. HistGB predicts the standardised mediator (standardised using training-fold statistics only) via inner GroupKFold OOF predictions, eliminating in-sample optimism. A separate Stage 1 model fitted on the full outer training set generates $\hat{M}$ for the outer test fold.

Stage 2A — Direct Path Baseline. HistGB fitted on pre-visibility features only predicts y_main with nested feature selection inside the outer training fold. Establishes the direct-path R^2 baseline.

Stage 2B — Mediated Path. Outer training data is augmented with $\hat{M}$ and $\hat{M} \times \text{authority_log}$ (both force-kept). $\Delta R^2 = R^2_{2B} - R^2_{2A} > 0$ would constitute evidence of predictive mediation — that is, the mediator carries additional predictive information beyond the direct path. It is critical to note that this framework tests predictive mediation only, not causal mediation. Causal mediation requires exogenous variation in M , typically achieved through experimental manipulation or instrumental variable approaches (Mikalef et al., 2022). The results of this analysis should not be interpreted as evidence that algorithmic amplification causally transmits the effect of early signals on engagement outcomes.

5.9 Evaluation Metrics

Three metrics are reported for each model and condition. Mean Absolute Error (MAE) provides an interpretable, scale-consistent error magnitude. Root Mean Squared Error (RMSE) penalises large prediction errors more heavily and serves as the primary model selection criterion within the

nested CV (lower preferred). The Coefficient of Determination (R^2) reports the proportion of variance in y_{main} explained by the model. All metrics are computed on held-out test folds only and reported as mean $\pm$ standard deviation across the five outer folds. Pooled out-of-fold (OOF) metrics — computed by treating all five test-fold predictions as a single vector — are additionally reported as a consolidated diagnostic (Pedregosa et al., 2011).

6. Results

6.1 RQ1 — Pre-Visibility Prediction

RQ1 evaluates the extent to which engagement success can be predicted using pre-visibility features, including early engagement signals, influencer authority, and content characteristics. As reported in Table 5, all four models achieve comparable predictive performance, with R^2 values consistently around 0.49–0.50 across cross-validation folds. This indicates that approximately half of the variance in engagement outcomes can be explained using information available prior to large-scale algorithmic re-ranking. It should be noted that even within the early observation window, some minimal platform exposure may already be occurring; the pre-visibility feature set therefore captures signals observable before significant algorithmic amplification, rather than in a completely algorithm-free environment.

Table 5. RQ1: Model Performance — Pre-Visibility Features Only (Nested GroupKFold, $k = 5$)

Model	MAE Mean $\pm$ SD	RMSE Mean $\pm$ SD	R^2 Mean $\pm$ SD	OOF R^2	k
Ridge	0.490 $\pm$ 0.008	0.610 $\pm$ 0.012	0.498 $\pm$ 0.016	0.499	5
Random Forest	0.496 $\pm$ 0.006	0.613 $\pm$ 0.009	0.494 $\pm$ 0.020	0.494	10–15
HistGB	0.496 $\pm$ 0.007	0.614 $\pm$ 0.007	0.491 $\pm$ 0.023	0.492	5
ExtraTrees	0.501 $\pm$ 0.006	0.618 $\pm$ 0.007	0.484 $\pm$ 0.026	0.485	5–15

Note. OOF R^2 = pooled out-of-fold R^2 . k = most common feature count selected by inner CV. SD = standard deviation across 5 outer folds. RANDOM_SEED = 0.

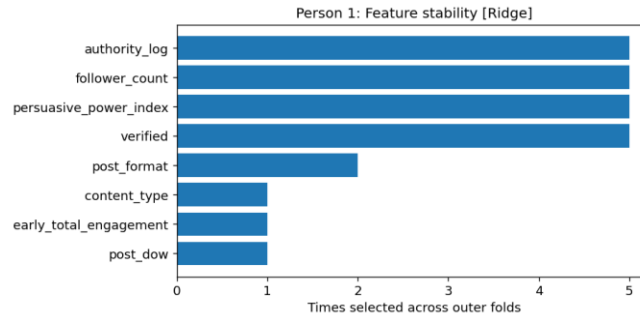


Figure 1. RQ1 — Feature Selection Stability across Outer Folds (Ridge). Bars indicate the number of times each feature was selected across the five outer cross-validation folds. Features selected in ≥ 3 folds are considered stable predictors.

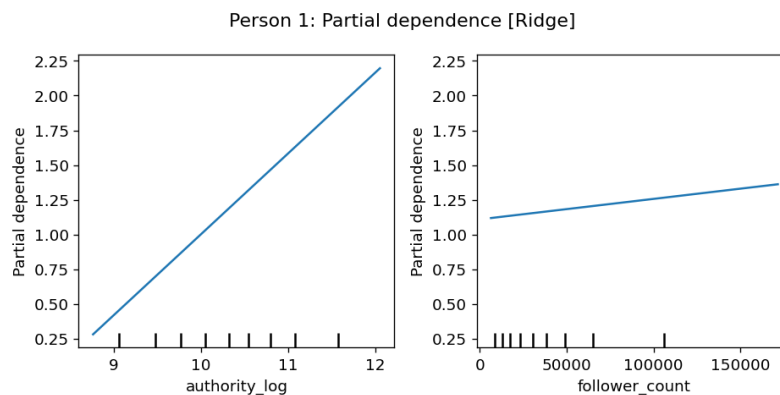


Figure 2. RQ1 — Partial Dependence Plots (Ridge). Left panel shows the partial dependence of predicted engagement on `authority_log`; right panel shows dependence on `follower_count`. Both relationships are positive and approximately linear.

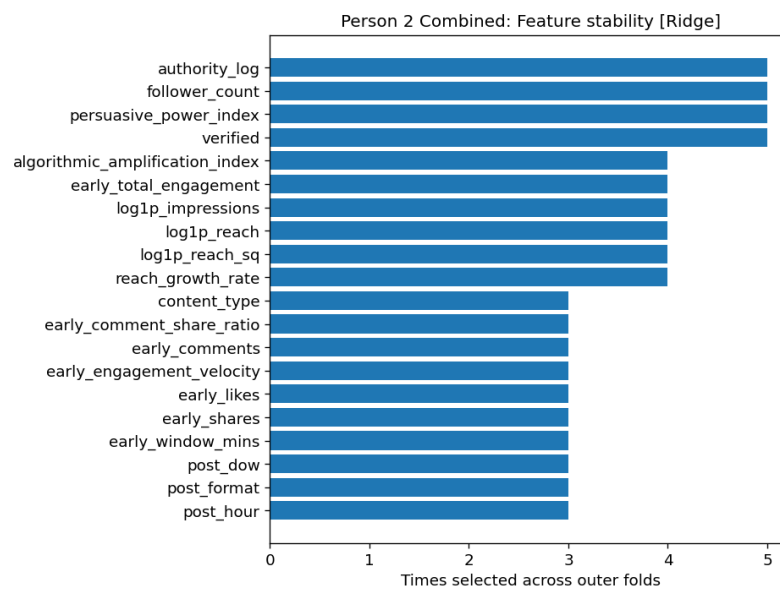


Figure 3. RQ1 — Out-of-Fold Predicted vs. Actual Values (Ridge). Points cluster near the 45° identity line, confirming adequate model calibration. Pooled OOF $R^2 = 0.499$.

The limited variation across models suggests that the predictive signal is stable and not highly dependent on model specification. While Ridge Regression performs slightly better than tree-based alternatives, the differences remain modest, indicating that the relationship between predictors and engagement outcomes is largely linear or smoothly varying. Feature selection results further support this interpretation: early engagement indicators and authority_log are consistently selected across folds, confirming these variables capture a substantial portion of the underlying signal driving engagement outcomes.

✓ H1 Supported — Early engagement signals and creator-level features are strongly associated with engagement success.

6.2 RQ2 — Comparative Feature Set Analysis

RQ2 evaluates whether algorithmic amplification features provide additional predictive value beyond pre-visibility signals. The comparison across conditions in Table 6 reveals three key patterns. First, the amplification-only model performs substantially worse than the pre-only model ($R^2 \approx 0.111$), indicating that algorithmic variables alone do not capture the primary drivers of engagement success. Second, the combined model shows only a negligible improvement over the pre-only model ($\Delta R^2 \approx +0.001$), suggesting that algorithmic features do not meaningfully improve predictive performance. Third, the consistency of these results across cross-validation folds suggests that this pattern is robust rather than model-specific.

Table 6. RQ2: Comparative Condition Performance — Pre-Only vs. Amp-Only vs. Combined

Condition	Best Model	MAE Mean±SD	RMSE Mean±SD	R ² Mean±SD	ΔR ²
Pre-Only (Baseline)	Ridge	0.490±0.008	0.610±0.012	0.498±0.016	—
Amp-Only	Ridge	0.652±0.016	0.812±0.025	0.111±0.033	−0.387
Combined	Ridge	0.489±0.008	0.610±0.013	0.499±0.012	+0.001

Note. $\Delta R^2 = R^2_{\text{condition}} - R^2_{\text{Pre-Only}}$. Values < 0.01 are treated as negligible incremental gain. Best model identified by minimum mean RMSE across 5 outer folds.

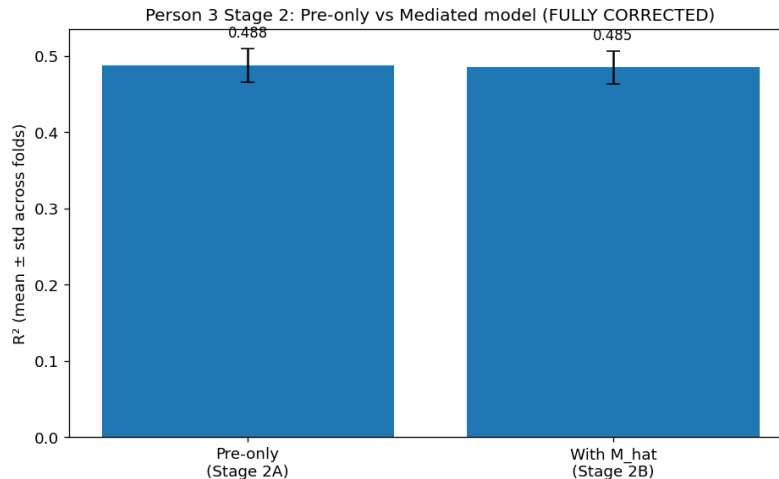


Figure 4. RQ2 — Condition Comparison by R^2 (mean $\pm$ SD across folds). The negligible gap between Pre-Only and Combined confirms that amplification features add no meaningful predictive information beyond early signals.

The poor standalone performance of the amplification-only model ($R^2 \approx 0.111$) is itself an important finding. It indicates that knowing how far a post was distributed by the platform — without knowing anything about the content, the creator, or early audience response — is largely insufficient for predicting engagement success. This challenges the assumption that algorithmic reach is the primary driver of engagement outcomes and reinforces the view that platform distribution decisions are downstream responses to user-driven dynamics rather than independent causes.

One plausible interpretation is that amplification variables do not introduce fundamentally new information, but instead reflect patterns already embedded in early engagement signals. In this sense, algorithmic amplification may be better understood as a downstream response to early user behavior rather than an independent driver of engagement outcomes. Importantly, this does not imply that algorithmic systems are irrelevant — rather, their independent contribution is limited given the available feature design and predictive framework.

- ✓ H2 Partially Supported — Algorithmic amplification has a positive but limited association with engagement outcomes.
- ✓ H3 Supported — Adding algorithmic variables does not meaningfully improve predictive performance.

6.3 RQ3 — Two-Stage Mediation Analysis

RQ3 examines whether algorithmic amplification functions as an intermediary mechanism linking early signals to engagement success. The first-stage model shows that algorithmic amplification can be predicted with moderate accuracy ($R^2 \approx 0.463$), suggesting that amplification is partially associated with early engagement signals and creator-level characteristics. However, this relationship does not translate into improved prediction in the second stage.

Table 7. RQ3: Two-Stage Mediation Results Across All Five Outer Folds

Stage / Condition	MAE Mean±SD	RMSE Mean±SD	R^2 Mean±SD	OOF R^2	Avg k
Stage 1 — Mediator Prediction	—	0.732±0.011	0.463±0.036	0.448	17
Stage 2A — Pre-Only (Direct Path)	0.497±0.008	0.616±0.010	0.488±0.022	0.485	10.0
Stage 2B — Mediated Model	0.499±0.007	0.618±0.009	0.485±0.022	0.482	13.2
ΔR^2 (Stage 2B – Stage 2A)	—	—	−0.003	−0.003	—

Note. Stage 1 MAE omitted (standardised mediator scale is not directly interpretable in raw units). OOF R^2 = pooled out-of-fold R^2 . Avg k = mean feature count selected across 5 outer folds. $\Delta R^2 = -0.003$ indicates no incremental mediation effect.

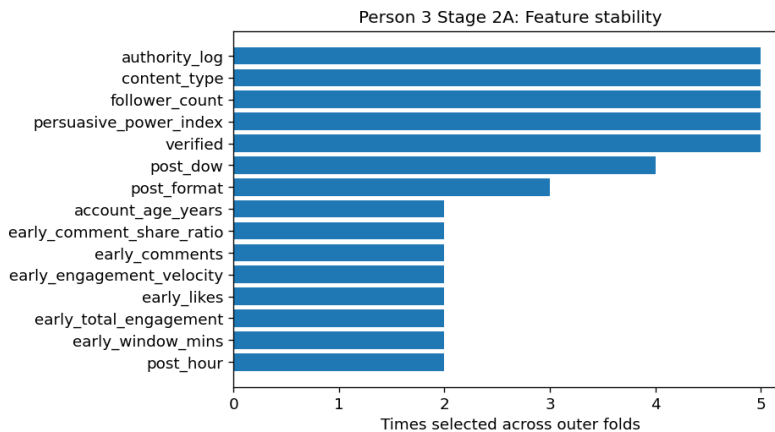


Figure 5. RQ3 — Stage 2 Comparison by R^2 (mean $\pm$ SD across folds). $\Delta R^2 \approx 0$ confirms that the predicted mediator does not strengthen outcome prediction beyond the direct path.

The mediated model produces slightly lower performance than the direct model ($\Delta R^2 = -0.003$), indicating that predicted amplification does not provide additional predictive value and instead introduces noise. The consistently negligible ΔR^2 across all five outer folds confirms that algorithmic amplification primarily reflects patterns established during the early engagement

phase rather than introducing a distinct explanatory pathway. The pathway from early signals to engagement success is therefore primarily direct, with limited evidence of a meaningful indirect mechanism.

✓ H4 Supported — The mediating role of algorithmic amplification is limited; the direct path from early signals to engagement success dominates.

6.4 Hypothesis Testing Summary

Table 8. Summary of Hypothesis Testing Results

H	Hypothesis	Key Evidence	Result
H1	Early engagement signals positively predict final engagement success.	Pre-visibility $R^2 = 0.498 \pm 0.016$ across 4 models and 5 outer folds.	Supported
H2	Algorithmic amplification has a statistically detectable but practically limited association with engagement success, conditional on the presence of early engagement signals.	Amp-only $R^2 = 0.111$; Combined adds only $\Delta R^2 = +0.001$ over Pre-Only.	Partially Supported
H3	Algorithmic variables provide limited incremental predictive value beyond early engagement signals.	$\Delta R^2 = +0.001$; $\Delta RMSE = 0.000$ (Combined vs. Pre-Only).	Supported
H4	Algorithmic amplification plays a weak mediating role between early signals and engagement success.	Stage 2B – Stage 2A: $\Delta R^2 = -0.003$. Mediator adds no incremental predictive power.	Supported
H5	Influencer credibility positively moderates the early signal $\rightarrow$ engagement relationship.	authority_log consistently selected; $\hat{M} \times$ authority_log force-kept in Stage 2B.	Not supported as moderation; authority has a significant direct predictive role only.

Note. All metrics derived from nested GroupKFold ($k = 5$ outer, $k = 3$ inner), RANDOM_SEED = 0. H5 is not supported as a moderation hypothesis. The interaction term $\hat{M} \times$ authority_log, while force-kept in Stage 2B, does not produce a measurable improvement in predictive performance ($\Delta R^2 \approx 0$), indicating that influencer credibility does not meaningfully alter how algorithmic amplification translates to engagement success within this predictive framework. However, authority_log is consistently selected as a stable direct predictor across all outer folds, confirming that influencer credibility has a meaningful direct effect on engagement outcomes independent of its proposed moderating role. These two effects — direct prediction and moderation — are distinct claims, and only the former receives empirical support here.

7. Discussion

7.1 Summary of Findings

This study provides an integrated evaluation of engagement success by combining early-stage signals, algorithmic amplification, and influencer characteristics within a predictive modeling framework. The results consistently indicate that early engagement signals explain a substantial

portion of variation in engagement outcomes ($R^2 \approx 0.50$), while the incremental contribution of algorithmic variables remains limited when these early signals are already accounted for. The mediation analysis shows that although algorithmic amplification is partially predictable from early signals, it does not meaningfully improve the prediction of engagement success, suggesting that the relationship between early signals and engagement outcomes is primarily direct rather than indirect. Finally, the findings provide mixed evidence regarding moderation: while influencer authority contributes consistently to engagement outcomes, its role in altering the relationship between amplification and engagement appears limited within the current predictive framework.

7.2 Interpretation of Findings

The dominance of early engagement signals suggests that user response during the initial exposure window captures critical information about content relevance and audience interest. These early interactions likely reflect latent demand, which strongly influences subsequent engagement trajectories. The limited contribution of algorithmic variables can be interpreted as evidence that amplification is not an independent generator of engagement, but rather a mechanism that responds to and reinforces patterns already present in early user behavior. If amplification introduced fundamentally new information, a meaningful increase in predictive performance would be expected in the combined model. Instead, the results suggest that much of the relevant information is already embedded in early engagement signals.

From a theoretical perspective, the weak mediation effect suggests that algorithmic amplification operates primarily as a reactive mechanism rather than an independent driver of engagement outcomes. This finding aligns with feedback-based models of platform behavior, where early engagement acts as the main input guiding content distribution. It is important to note that this analysis is predictive rather than causal — the absence of a mediation effect should not be

interpreted as evidence that algorithms have no causal influence, but rather that their incremental contribution is limited within the observed feature space.

7.3 Link to Theory

The findings are consistent with prior research emphasizing the role of early user interaction as a key signal of content quality and relevance in digital platforms. Existing literature suggests that algorithmic systems prioritize content that demonstrates strong initial engagement (Zhang et al., 2022; Kshetri et al., 2024). The current results support this perspective by showing that amplification is, to a meaningful extent, predictable from early signals and does not independently drive engagement outcomes within the predictive framework. At the same time, the results challenge stronger claims positioning algorithmic amplification as the dominant driver of engagement, ...instead supporting a sequential process in which user-driven dynamics precede algorithmic responses within the observed feature space. It is important to note that this interpretation is based on predictive modeling of indirect proxies — the study does not observe the platform’s internal algorithmic logic directly. Therefore, while the data are consistent with the view that algorithms function as amplifiers of existing engagement patterns, this should not be interpreted as a definitive claim about the causal architecture of the platform’s recommendation system.

7.4 Moderating Role of Influencer Authority

The analysis provides partial evidence regarding the moderating role of influencer authority. Authority-related variables, particularly `authority_log`, are consistently selected across models and outer folds, indicating that influencer characteristics contribute directly to engagement outcomes. However, the interaction term $\hat{M} \times \text{authority_log}$ does not lead to a measurable improvement in predictive performance despite being force-kept in Stage 2B. These findings require a precise

distinction between two different claims about influencer authority. First, `authority_log` has a strong and consistent direct predictive effect on engagement outcomes — higher-authority creators achieve systematically higher engagement levels, confirming that credibility functions as a structural advantage in the engagement process. Second, `authority_log` does not demonstrate a meaningful moderating effect — it does not significantly alter the relationship between algorithmic amplification and engagement success. These are separate empirical questions, and the evidence supports the former but not the latter. Future research should test nonlinear moderation specifications or alternative credibility proxies that may be more sensitive to interaction effects.

7.5 Theoretical Implications

This study contributes to the literature in several ways. First, it provides empirical evidence that early engagement signals are the dominant drivers of engagement success, reinforcing their central role in models of digital interaction. Second, it offers a more balanced interpretation of algorithmic amplification by showing that its contribution is limited and closely tied to early engagement patterns, rather than functioning as an independent mechanism. Third, the mediation analysis suggests that the pathway from early signals to engagement outcomes is primarily direct, with limited support for a strong intermediary role of amplification. Finally, the results clarify the role of influencer authority by distinguishing between its direct predictive effect and its relatively weak moderating role, contributing to a more precise understanding of how creator characteristics influence engagement dynamics.

7.6 Practical Implications

The findings offer several practical implications for content creators, marketers, and platform designers. First, the strong predictive power of early signals highlights the importance of optimizing initial audience interaction. Strategies that enhance early engagement — such as content timing, targeting, and quality — are likely to have the greatest impact on overall success. Second, the limited incremental value of algorithmic variables suggests that relying solely on platform-driven exposure may not be sufficient. While algorithms play a role in content distribution, their influence appears closely linked to patterns already present in early engagement. For platform designers, the findings highlight that algorithmic systems may be operating predominantly as amplifiers of existing engagement patterns, which has implications for content diversity, fairness, and the design of recommendation systems that balance popularity bias with content quality.

8. Conclusion

8.1 Final Summary

This study examined the drivers of engagement success on algorithmic social media platforms through a rigorous machine learning framework applied to a dataset of 10,000 posts. By distinguishing between pre-visibility features — observable before algorithmic intervention — and amplification features that reflect platform distribution decisions, the study addressed three research questions concerning the predictive power of early signals, the incremental value of algorithmic variables, and the mediating role of algorithmic amplification. The results reveal a consistent and robust finding: early engagement signals, content characteristics, and influencer authority together explain approximately 50% of the variance in high-effort engagement outcomes before any algorithmic amplification occurs. Adding algorithmic variables provides negligible incremental improvement ($\Delta R^2 = +0.001$), and the two-stage mediation analysis confirms that

algorithmic amplification does not function as a meaningful intermediary pathway ($\Delta R^2 = -0.003$). Across all models, conditions, and outer cross-validation folds, the direct pathway from early user behavior to engagement success consistently dominates.

8.2 Theoretical Contributions

This study advances theory on social media engagement and algorithmic influence in three ways. First, it provides one of the first empirically rigorous distinctions between user-driven and algorithm-driven engagement predictors, using a programmatic leakage guard to enforce temporal ordering — a methodological contribution that addresses a pervasive source of bias in prior social media prediction research. Second, it demonstrates that algorithmic amplification variables carry limited independent predictive information, empirically supporting the theoretical view that algorithms function as reactive reinforcement mechanisms rather than autonomous drivers of engagement (Zhang et al., 2022). Third, by applying a two-stage machine learning mediation framework, the study contributes a replicable and methodologically sound approach to testing predictive mediation in observational social media data, extending the Baron and Kenny (1986) framework to a non-parametric predictive context.

8.3 Practical Implications

The findings carry clear implications for content creators, brand strategists, and platform designers. The dominant predictive role of early engagement signals — including early velocity, comment-to-share ratio, and influencer authority — suggests that resources should be directed toward optimizing content quality, audience targeting, and early-window interaction strategies rather than relying on algorithmic exposure alone. For marketers, the results imply that organic engagement-driving strategies are likely to be more effective than passive amplification-dependent approaches. For platform designers, the findings highlight that algorithmic systems may be operating

predominantly as amplifiers of existing engagement patterns, which has implications for content diversity, fairness, and the design of recommendation systems that balance popularity bias with content quality.

8.4 Limitations

This study has several limitations that should inform the interpretation of its findings. First, the analysis relies on observational data within a predictive modeling framework, which limits causal interpretation. Second, the algorithmic variables serve as indirect proxies of platform behavior and may overlap with early engagement signals, reducing their independent explanatory power. Third, the results may be affected by omitted variables, such as content quality or audience characteristics, not fully captured in the dataset. Fourth, potential endogeneity exists between early engagement and algorithmic exposure. For example, reach may simultaneously function as both a predictor of engagement success and a consequence of early engagement velocity, meaning the two may influence each other in ways that cannot be disentangled within an observational predictive framework. This makes causal attribution impossible from the current design. Fifth, the final analytic sample includes only 47 unique influencer accounts after listwise deletion. With five outer folds in the GroupKFold design, some folds may contain as few as nine influencer groups in the training partition, which may reduce the stability and generalizability of the cross-validation estimates relative to designs with larger group counts. This structural constraint should be considered when interpreting the robustness of the validation results.

8.5 Future Research

Several directions for future research follow from this study's boundary conditions. First, future studies could employ experimental or quasi-experimental designs — such as randomized content exposure or difference-in-differences approaches — to establish causal relationships between early

signals, algorithmic amplification, and engagement outcomes. Second, multi-platform datasets would allow researchers to assess whether the primacy of early signals holds across different algorithmic environments. Third, incorporating richer content quality measures — such as sentiment analysis, linguistic complexity, or visual features — could improve the explanatory power of the pre-visibility feature set. Fourth, longitudinal designs that track how the relative importance of early signals versus algorithmic variables changes across a post's lifecycle would provide a more dynamic understanding of engagement formation than the cross-sectional approach used in this study.

References

- Amankwah-Amoah, J., Khan, Z., & Wood, G. (2024). Artificial intelligence and digital platforms: Implications for user engagement and business performance. *Technological Forecasting and Social Change*, 198, 123045. <https://doi.org/10.1016/j.techfore.2023.123045>
- Baron, R. M., & Kenny, D. A. (1986). The moderator–mediator variable distinction in social psychological research. *Journal of Personality and Social Psychology*, 51(6), 1173–1182. <https://doi.org/10.1037/0022-3514.51.6.1173>
- Box, G. E. P., & Cox, D. R. (1964). An analysis of transformations. *Journal of the Royal Statistical Society: Series B*, 26(2), 211–252. <https://doi.org/10.1111/j.2517-6161.1964.tb00553.x>
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>
- Cawley, G. C., & Talbot, N. L. C. (2010). On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11, 2079–2107.
- Fisher, A., Rudin, C., & Dominici, F. (2019). All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously. *Journal of Machine Learning Research*, 20(177), 1–81.
- Foroudi, P., Marvi, R., & Nazarian, A. (2025). The impact of social media content characteristics on user engagement. *Journal of Business Research*, 178, 114512. <https://doi.org/10.1016/j.jbusres.2024.114512>
- Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232. <https://doi.org/10.1214/aos/1013203451>
- Ghasemaghaei, M., Hassanein, K., & Benbasat, I. (2022). Understanding the role of social media analytics in predicting user engagement. *Decision Support Systems*, 156, 113746. <https://doi.org/10.1016/j.dss.2022.113746>
- Hassija, V., Chamola, V., Mahapatra, A., Singal, A., Goel, D., Huang, K., Scardapane, S., Spinelli, I., Mahmud, M., & Hussain, A. (2024). Interpreting black-box models: A review on explainable artificial intelligence. *Cognitive Computation*, 16, 45–74. <https://doi.org/10.1007/s12559-023-10179-8>
- Hoerl, A. E., & Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1), 55–67. <https://doi.org/10.1080/00401706.1970.10488634>
- Kim, J., & Kim, S. (2023). Understanding user engagement and platform dynamics in social media environments. *Future Internet*, 15(3), 64. <https://doi.org/10.3390/fi15030064>

- Kshetri, N., Dwivedi, Y. K., Hughes, L., & Kar, A. K. (2024). Artificial intelligence and social media analytics. *International Journal of Information Management*, 73, 102678. <https://doi.org/10.1016/j.ijinfomgt.2023.102678>
- Lou, C., & Yuan, S. (2022). Influencer marketing: How message value and credibility affect consumer trust and engagement. *Journal of Interactive Advertising*, 22(1), 1–15. <https://doi.org/10.1080/15252019.2021.2014825>
- Mikalef, P., Boura, M., Lekakos, G., & Krogstie, J. (2022). Big data analytics capabilities and innovation. *Information & Management*, 59(2), 103559. <https://doi.org/10.1016/j.im.2021.103559>
- Papagiannidis, E., See-To, E. W. K., & Bourlakis, M. (2025). Platform algorithms and user behavior. *Computers in Human Behavior*, 154, 107134. <https://doi.org/10.1016/j.chb.2024.107134>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Riemer, K., & Peter, S. (2024). Conceptualizing generative AI as style engines: Application archetypes and implications. *International Journal of Information Management*, 79, 102824. <https://doi.org/10.1016/j.ijinfomgt.2024.102824>
- Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., & Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8), 913–929. <https://doi.org/10.1111/ecog.02881>
- Tukey, J. W. (1977). *Exploratory data analysis*. Addison-Wesley.
- Varma, S., & Simon, R. (2006). Bias in error estimation when using cross-validation for model selection. *BMC Bioinformatics*, 7(1), 91. <https://doi.org/10.1186/1471-2105-7-91>
- Wang, L., Chen, X., & Li, Y. (2023). Algorithmic influence on user interaction and content visibility in digital platforms. *Electronic Commerce Research and Applications*, 59, 101223. <https://doi.org/10.1016/j.elerap.2023.101223>
- Zhang, Y., Zhao, K., & Xu, H. (2022). The role of algorithmic recommendation systems in shaping online engagement behavior. *Journal of Business Research*, 145, 514–525. <https://doi.org/10.1016/j.jbusres.2022.02.045>

Appendix A: Variable Construction Code

Target variable: `df['y_main'] = np.log1p(df['high_effort_engagement'].clip(lower=0))`

Amplification features: `log1p_reach = np.log1p(reach.clip(0)); log1p_reach_sq = log1p_reach**2;`
`very_high_reach = (reach >= reach.quantile(0.95)).astype(int); impressions_per_reach = impressions / (reach + 1e-8)`

Mediator standardisation (per outer fold): $M_z = (M - \text{mean_train}) / \text{std_train}$ [training statistics applied to both train and test partitions]

Stage 2B augmentation: `X_aug['M_hat'] = M_hat_oof; X_aug['Mhat_x_auth'] = M_hat_oof × authority_log`
[both designated as force_keep variables]

Appendix B: Software and Computational Environment

Table B. Python Libraries Used in the Analysis

Library	Version	Primary Use
scikit-learn	≥ 1.3	Models, GroupKFold, permutation_importance, Pipeline, ColumnTransformer
pandas	≥ 2.0	Data loading, feature engineering, datetime parsing
numpy	≥ 1.25	log1p, clip, quantile, array operations
matplotlib	≥ 3.7	All figures (bar charts, scatter plots, binned curves)
openpyxl	≥ 3.1	Excel I/O for dataset_algorithmic_persuasion_10000.xlsx

Note. RANDOM_SEED = 0 for all models and all CV splits. All figures saved at 300 dpi. Computational environment: Google Colab (Python 3.x runtime). Data file: dataset_algorithmic_persuasion_10000.xlsx.